

Evaluating the effectiveness of supplemental instruction using a multivariable analytic approach

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ABSTRACT

Supplemental instruction (SI) is a widely implemented peer-led academic support program. Assessments of SI's effectiveness are often confounded by student ability, achievement, motivation and demographic characteristics, and frequently limited by available resources. We modelled the impact of SI attendance on the grades, pass rates and retention of over 4000 university students. Percentage grade increases of up to 10.4% were associated with a full semester of SI attendance, and largest for males, younger students and those who had not previously engaged with the program. Furthermore, pass rates were 10% higher for full attendees. Finally, full attendance was associated with a 3% and 9% increase in the probability of reenrolling the following year for students who had and had not previously engaged, respectively. These estimates were calculated controlling for measures of ability, achievement and motivation, plus several demographic factors. Our methods can be applied to evaluate any opt-in academic support program.

Supplemental Instruction (SI; also known as Peer Assisted Study Sessions [PASS] and Peer Assisted Learning [PAL]) emerged as an academic support intervention at the University of Missouri-Kansas City in the early 1970s (International Center for Supplemental Instruction, 2021; Martin, 2008). Since that time, SI has been deployed in over 1500 tertiary institutions in over 29 countries (Martin, 2008), and currently supports the learning of hundreds of thousands of students at an estimated cost of tens of millions of dollars annually (Dawson, van der Meer, Skalicky, & Cowley, 2014). SI is commonly attached to traditionally difficult, 'high-risk' subjects, and runs parallel to the standard curriculum (Martin & Arendale, 1992). SI sessions are not remedial, but are optional and open to all students, regardless of pre-existing ability (Martin & Arendale, 1992). The peer educators who facilitate these sessions are academically successful higher year students with good communication skills. They do not re-teach, nor work on assessable material or mark students' work (Australasian Centre for PASS, 2010).

Rather, they design and deliver learning activities and facilitate discussions that promote a deeper understanding of subject content and the development of transferable study skills (Australasian Centre for PASS, 2017; Dawson et al., 2014; International Center for Supplemental Instruction, 2021). Peer educators/facilitators are supported in their role by trained and accredited SI supervisors, who are themselves supported by a national SI accrediting center, and ultimately the International Center for Supplemental Instruction (Dawson et al., 2014).

As would be expected for such a widely implemented academic support program, SI has attracted a substantial body of research examining its effectiveness. Early work by Martin & Arendale, 1992, who aggregated data from 49 US universities, indicated that SI attendees achieved higher grades and were more likely to 'succeed' (i.e., earn fewer D and F grades, and withdraw at lower rates) than non-attendees. Smaller datasets from the University of Missouri-Kansas City also suggested that these findings persisted across different levels of prior

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achievement, and that SI attendees were more likely to reenroll, and ultimately graduate, than non-attendees. [Martin & Arendale, 1992](#) early findings - that SI attendance is positively correlated with grades, retention and graduation, and negatively correlated with failure and withdrawal - have been broadly replicated many times since ([Dawson et al., 2014](#)). For example, in a recent systematic review of SI research ([Dawson et al., 2014](#)), a consistent positive link between attendance and grades was identified.

Nonetheless, despite this body of research supporting SI as an effective learning intervention, the existing evidence has been critiqued on a number of grounds. First, there has been a tendency to assess the effectiveness of SI in just one subject, or a small number of subjects. For example, of the 29 studies included in [Dawson and colleagues' \(2014\)](#) review, 19 studies sampled from one subject, and only two sampled from more than four. Therefore, given that learning interventions may work differently across various study disciplines (with heterogeneous content), we identified a need to investigate the benefits of SI across a large number of subjects and courses. Second, despite [McCarthy, Smuts, and Cosser's \(1997\)](#) early advice against dichotomizing attendance (e.g., 'attended' versus 'did not attend'), which results in unnecessary information loss, reduced statistical power and a range of other undesired outcomes (c.f., [MacCallum, Zhang, Preacher, & Rucker, 2002](#)), this practice has remained prevalent in the SI literature. Furthermore, the definition of an attendee has varied considerably across studies. For example, in those studies reviewed by [Dawson et al. \(2014\)](#), an attendee was variously defined as a student who attended 1, 3, 5, 12, or an unspecified number of SI sessions. In five studies, several 'levels' of attendance were created, but in only two studies ([Cheng & Walters, 2009](#); [Dancer, Morrison, & Smith, 2007](#)) was attendance analyzed as a discrete variable. In the current study, we wanted to ensure that the inherently continuous nature of attendance was retained. Third, there exists a need to incorporate prior ability, achievement, motivation and other potentially confounding variables into analyses ([Dancer, Morrison, & Tarr, 2015](#); [Dawson et al., 2014](#); [Paloyo, 2015](#)). Amongst the studies included in [Dawson and colleagues' \(2014\)](#) review, the most common control variables were varied indicators of pre-tertiary ability (e.g., high-school GPA, university/college entrance scores; 20 studies), tertiary achievement prior to SI (five studies) and self-reported motivation (three studies). However, only six studies considered two of these factors, and none considered all three. In the current study we controlled pre-tertiary ability, as well as tertiary achievement prior to SI, a proxy for motivation and several demographic variables. As a measure of pre-tertiary ability, we used the Australian Tertiary Admission Rank (ATAR). As measures of prior achievement we used students' prior semester weighted averages and whether or not they had ever been placed on 'conditional status' by an exam board. At Curtin University, where we collected the data reported in this paper, being placed on conditional status is typically triggered by failing one or more units in the previous semester. As a proxy for motivation, we used students' prior engagement with the SI program, relative to opportunity. The demographic variables included age, gender, number of parents with a tertiary qualification, whether or not the student identified as an Indigenous Australian, and whether they were from a postcode designated as regional, remote or low socioeconomic status. Like all the variables in this study, these were extracted from university records maintained for varied operational purposes.

One further limitation of the existing body of research relates to the fact that large-scale randomized controlled trials (RCTs) are quite rare within education research (due to the ethical and practical difficulties of implementation, and the large focus on individual development models driven by constructivist theories). As a result, it is difficult to control for all variables, and calls for 'quasi-experimental' approaches have recently been made (e.g., [Paloyo, 2015](#)). Ultimately, these suggestions have led to some scholars implementing research designs that are pragmatic rather than strict RCTs (e.g., [Dancer et al., 2015](#); [Musah & Ford, 2017](#); [Paloyo, Rogan, & Siminski, 2016](#)). For example, in these

studies, nuanced statistical modelling has been used in an attempt to control for potentially confounding variables (e.g., students' ability and demographic characteristics). Overall, through their unique designs, these studies have generally found that SI is beneficial for students' educational trajectories (particularly for those in 'at-risk' groups; e.g., international students and lower-achieving students; [Dancer et al., 2015](#)). Furthermore, in a similar fashion, [Paloyo et al. \(2016\)](#) aimed to control for self-selection bias by using a "randomized-encouragement-design" experiment (i.e., a modified version of an RCT, which draws upon differing levels of incentives for participants within various subgroups, in order to produce groups with different attendance levels). Through this unique approach, [Paloyo and colleagues](#) observed a link between SI and students' grades. Overall, while these studies utilized relatively novel designs to optimize data quality, they were still limited in various ways (e.g., lacking a fully randomized model, with an inability to control for all variables). Nonetheless, these recent investigations have certainly provided insights relating to how innovative statistical mechanisms and research designs can be used to more adequately explore SI's effectiveness.

From this emerging body of work, it appears that recent advances in statistical methods and software have provided the tools and expertise to investigate the effectiveness of SI programs using advanced modelling techniques. These approaches have opened up new possibilities for understanding factors associated with effective learning. Additionally, prospective studies can be costly, resource intensive, and not ideally suited to providing timely advice for use in decision making, as might be required during the early stages of implementing a SI program. In these cases, and especially where there are available datasets on enrolments, academic performance, retention and SI attendance, a multivariable analysis could prove useful in estimating program effectiveness. Initial estimates could be used to directly inform the decision analyses conducted for subsequent institutional planning, or to adapt or tailor aspects of the program to result in improved performance.

The present study used multivariable mixed models to analyze large existing datasets to assess the effectiveness of a university-wide accredited SI program. Specifically, our goals were to determine the extent to which SI attendance was associated with subject grades, subject pass rates and retention into the next academic year. Relative to achievement, retention is an understudied variable in the SI literature ([Dawson et al., 2014](#)). However, given its funding implications, it is a large concern for universities ([DeBerard, Spielmans, & Julka, 2004](#); [Thammasiri, Delen, Meesad, & Kasap, 2014](#)). Recent work suggests that the impact of SI on retention may be as impressive as its impact on achievement. For example, in a large, multi-year study of several SI supported subjects, [van der Meer, Wass, Scott, and Kokaua \(2017\)](#) observed that students who engaged with the SI program in first year were considerably more likely to reenroll into the second year of their course, as well as more likely to graduate within six years of starting their course. These effects were estimated after controlling pre-tertiary ability, concurrent achievement, school decile (as a proxy for socio-economic status) and current living circumstances. However, [van der Meer and colleagues](#) did not include any measure of motivation, nor account for the hierarchical structure of their data in their analyses. In the current study we do both, and in doing so provide an example of how contemporary mixed modelling techniques may be used to assess the effectiveness of any voluntary academic support program.

1. Method

1.1. Study context

Curtin University is an Australian non-residential metropolitan university with students commuting daily to lectures and tutorials. Establishing an accredited SI program at Curtin University in 2012 was part of wider strategy to improve student retention, especially for students from less represented socio-economic backgrounds. This program is named

UniPASS. SI program accreditation (as obtained by UniPASS) requires adherence to four broad core components: (i) a specifically trained program supervisor; (ii) formal training of facilitators; (iii) session planning; and (iv) that facilitators must attend at least 60% of lectures for the supported unit ([International Center for Supplemental Instruction, 2021](#)).

In 2014, the most recent year for which data were available, UniPASS supported 40 subjects. These were first (70%) or second (30%) year subjects spanning disciplines as diverse as accounting, biology, chemistry, economics, geology, mathematics, pharmacology, psychology and software engineering. In 2014 there were 9152 students enrolled in supported subjects, of whom 2124 attended at least one UniPASS session. Sessions have an ideal ratio of 20 participants to one facilitator, although participant numbers may increase near exam periods. UniPASS sessions are held weekly in dedicated classrooms and generally run for 50 min ([Curtin University, 2017](#)). New UniPASS facilitators are required to complete 27 hours of online and face-to-face training prior to commencing UniPASS facilitation ([Allen, Tonta, Pereira, & Roberts, 2019](#)). Facilitator skills include the ability to run a range of learning activities, efficient instruction delivery, concept checking, managing group dynamics, redirecting questions, Socratic questioning, minimizing hierarchy, participant-developed content and formal session planning ([Allen et al., 2019](#)).

1.2. Participants

Data for at least 4140 students, in conjunction with at least 5797 recorded attendance points, were analyzed. Sample sizes varied slightly for each analysis because data for some variables were not available for all students. Data for only those students who were enrolled in a UniPASS supported subject at Curtin University during 2012–2014, and had previously studied a UniPASS supported subject, were analyzed. The latter constraint was necessary for assessing effects of prior engagement. Furthermore, data for only those students with an Australian tertiary entrance exam score (ATAR) were analyzed because *a priori* this was considered an important measure of prior ability to include in analyses. As a consequence of this decision, the sample only included a very small number of international students (2.4%), despite international students comprising a considerably large proportion (13.5%) of the population of the supported subjects during the data collection period. The demographic characteristics of the sample for each analysis are reported in [Table 1](#).

1.3. Datasets

Three different datasets were merged for this study. “Administration” data were extracted from the university student records database. These data were comprised of student grades for SI-supported subjects completed during 2012–2014, whether students passed or failed these subjects, and several demographic variables. “Attendance” data were obtained from the SI program managers. These data were comprised of the number of SI sessions attended by each student for each subject (ATTEND). A third data source indicating whether each student’s enrolment continued into the subsequent year or not was also used (the “Retention” dataset). The retention dataset was compiled annually from Curtin University enrolment records by the University’s Office for Strategy and Planning.

Outcome measures were students’ percentage marks (0–100%) for SI-supported subjects (grade), whether students passed or failed each subject (status) and whether or not students’ enrolments continued into the next year (retention). It should be noted that this definition of a retained student is consistent with the [Australian Government Department of Education and Training’s \(2017\)](#) definition: a student “who commenced a bachelor course in year(x) and did not complete in year (x), and continued in year(x+1)”. The effects of ATTEND on grade ($N = 5797$) and status ($N = 5840$) were assessed using the merged

Table 1

Summary statistics for each analysis. Level 1 predictors (L1) are variables with values that vary for individual students. Level 2 predictors (L2) are variables that do not vary for individual students. Means and standard deviations (sd) are reported for continuous variables and percentages (%) and sample sizes (n) are reported for selected categories of categorical variables. Refer to Appendix for variable definitions.

		Analysis		
Characteristic	Group	grade	status	retained
<i>SI measurements</i>		Mean (sd)	% (n)	% (n)
Grade		59.20 (16.89)		
Status	Pass		78.8 (4603)	
Retained	Yes			88.7 (5198)
<i>L1 predictors</i>		<i>N = 5797</i>	<i>N = 5840</i>	<i>N = 5863</i>
		Mean (sd)	Mean (sd)	Mean (sd)
ATTEND		0.74 (2.28)	0.73 (2.28)	0.73 (2.27)
P_OPPORT		0.81 (2.11)	0.81 (2.11)	0.80 (2.11)
SEM_NUM		2.57 (0.95)	2.56 (0.94)	2.57 (0.95)
SPWA		65.76 (13.88)	65.60 (13.98)	65.32 (14.60)
		% (n)	% (n)	% (n)
P_CONDTIT	Yes	22.9 (1326)	22.9 (1338)	23.2 (1358)
P_ENGAGE	Yes	22.5 (1302)	22.5 (1312)	22.4 (1312)
<i>L2 predictors</i>		<i>N = 4140</i>	<i>N = 4167</i>	<i>N = 4174</i>
		Mean (sd)	Mean (sd)	Mean (sd)
AGE		21.67 (4.68)	21.68 (4.70)	21.70 (4.71)
ATAR		81.40 (8.17)	81.39 (8.17)	81.38 (8.17)
		% (n)	% (n)	% (n)
GENDER	Male	41.3 (1709)	41.6 (1734)	41.7 (1740)
INDIGENOUS	Yes	0.6 (25)	0.6 (25)	0.6 (25)
SES	Low	11.5 (476)	11.5 (479)	11.5 (480)
PARENT	0	54.9 (2273)	54.9 (2288)	54.9 (2293)
PARENT	1	28.6 (1182)	28.6 (1191)	28.6 (1194)
PARENT	2	16.5 (685)	16.5 (688)	16.5 (687)
REGIONAL	Yes	9.2 (382)	9.3 (386)	9.3 (387)
REMOTE	Yes	1.4 (57)	1.4 (59)	1.4 (59)

Administration and Attendance data. The effect of ATTEND on retention ($N = 5863$) was assessed using the Administration and Attendance data merged with the Retention data.

Some further restrictions were required to exclude records with missing values (e.g., missing ATAR scores) and zero grades (for the analysis of grade). Zero grades identified students that had either failed to submit any assessments or scored zero for all submitted assessments. It is likely that some, if not all, of these students simply did not engage with their subjects (1.2% of cases). The inclusion of zero grades also markedly affected the skew of standardized residuals and the distribution of leverage (Cook’s Distance) scores for models fitted to grade.

These data were hierarchical. Variables with values varying only between students were Level 2 variables. These included students’ ATAR scores and several demographic variables (AGE, INDIGENOUS, GENDER, SES, PARENT, REGIONAL, REMOTE; see the Appendix for the list of variable names and definitions). Variables with values varying only among groups of students, which identified the clustering of students enrolled within different subjects (UNIT) and courses (COURSE), were Level 3 variables. All other variables, whose values varied within groupings of student (STUDENT_ID), semester, year, and UNIT, were Level 1 variables. These included variables quantifying prior university achievement (SPWA, P_CONDTIT), prior experience (P_ENGAGE, SEM_NUM), prior motivation (P_OPPORT) and ATTEND. Summary statistics are presented in [Table 1](#).

The SEM_NUM variable, which is the number of semesters enrolled since the student first studied a SI subject, was included to adjust for the effect of pooling data across semesters and years. Year was not included, as it would have been aliased with SEM_NUM. Excluding Year from analyses assumed a negligible influence on results from variation attributed to different cohorts over time. This decision was supported by the results of a preliminary analysis, which suggested that Year, as a main effect, was unimportant. Furthermore, effects from cumulative attendance were not tracked through time within semesters for each

student but instead were aggregated into a total attendance figure for each student for each semester. This was not an analysis decision inasmuch as it was a limitation of the available data, since measures of student learning outcomes within semesters were not available. Therefore, analyses presented in this study estimate the average effects of attendance on learning outcomes for the student population enrolled in SI-supported subjects at Curtin University from 2012 to 2014, from cross-sectional data.

1.4. Analysis

A separate analysis was conducted for each academic outcome (student grade, pass/fail status, or retention). Each analysis involved fitting a statistical model to estimate the association of SI attendance (the model predictor representing the “exposure” = number of sessions attended) with academic outcome (the dependent or response variable), after controlling for other factors, representing student ability, achievement, motivation, and demographic characteristics (model predictors). Model selection was performed, by dropping non-significant fixed and random effect terms. Hypothesis tests were then done using the final model to assess if SI attendance, or SI attendance interaction terms, were independently associated with the academic outcome.

The statistical models used were linear mixed models (LMMs) and generalized linear mixed models (GLMMs). More complex hierarchical models were initially considered, to explicitly account for the clustering of related observations within levels of STUDENT_ID, UNIT and COURSE. However, these were not pursued because a relatively high percentage (69.6%) of students had one observation only. Accordingly, the majority of student responses would have had a sample variance of zero and variation among student responses would have been mostly reflective of the variation for Level 1 variables in this more complex setting. Therefore, LMMs and GLMMs, incorporating random effects for COURSE and UNIT, were used. Levels of COURSE and UNIT were modelled as random intercepts to account for their suspected effect(s) on the distribution of residuals for closely related observations (i.e., for the same course or unit). COURSE and UNIT were treated as partially-crossed factors because students that were enrolled in different courses could study the same subjects.

Furthermore, the type of statistical model varied according to the type of outcome variable and general properties of the model residuals. A LMM was used to model transformed grades and binominal GLMMs with logit link function was used to model binary responses for status (i.e., pass or fail) and retention (i.e., retained or not retained). Box-Cox power transformations of a LMM for grade, with all candidate model predictors included, identified that the log-likelihood was maximized for $\lambda = 1.53$. As the 95% confidence interval for this λ (1.47, 1.59) bracketed 1.5, a LMM was fitted to the transformed square root of the cubed grades ($\text{grade}^{3/2}$). To address heteroskedasticity in residual variance for the LMM, residual variance was modelled as a power function of the fitted values ($\text{var}[e_i] = \sigma^2|\hat{y}_i|^{2\delta}$). To account for non-linearity, SPWA was transformed firstly by dividing by 100, and then modelling as a degree-two polynomial ($\text{pol}(\text{SPWA}, 2)$). Model selection followed the “top-down” approach of West, Welch, and Galecki (2007; refer to Supplementary Material, S1 for further details).

In each analysis, the final selected statistical model was used to describe and test the association of SI attendance with academic outcome. Likelihood ratio tests comparing nested models with and without ATTEND (or the retained ATTEND interaction term) assessed the significance of the association of SI attendance with outcome, after controlling for other model predictors (West et al., 2007; refer to Supplementary Material, S1 for further details). Plots and tables showing the direction and size of effects were constructed using back-transformed model-predicted values (estimated marginal means) from the best-fitting models. The net change in predicted value from attending the full complement of SI sessions held for each subject (full attendance),

as compared to zero attendance (not exposed), was chosen as a measure of effect size for all analyses. As almost all (97.6%) of attendance data were for ≤ 11 sessions per student per unit per semester, the number of sessions for full exposure to SI was selected to be 11.

Analyses were conducted in R version 3.2.3 (R Core Team, 2015), using the *lme4* and *nlme* packages for model fitting, as well as the *car*, *effects*, and *influence.ME* packages for model testing, predictions, and assessing diagnostics of fit (Bates & Martin, 2015; Fox, 2003; Fox & Weisberg, 2011; Nieuwenhuis, te Grotenhuis, & Pelzer, 2012; Pinheiro, Bates, DebRoy, Sarkar, & R Core Team, 2015).

2. Results

2.1. Grade: LMM 1

Retained predictor variables in the final selected model (LMM 1) for $\text{grade}^{3/2}$ and estimated variance components are shown in Table 2. All proxies for prior achievement (SPWA, P_CONDIT) and motivation (P_OPPORT) as main effects, and the interactions of ATTEND $\times$ P_ENGAGE, ATTEND $\times$ GENDER and ATTEND $\times$ AGE, were retained in the final model. The main effect of SEM_NUM was also retained.

Table 2

Summary statistics, tests of fixed effects for model selection, and estimated variance components in final models. * = *P*-values marked as **bold** signify that this term was retained in the final selected model; NE = not estimated due to convergence problems; ** = calculated using fixed dispersion of $\pi^2/3$ (Snijders & Bosker, 1999).

Model:	LMM 1	GLMM 1	GLMM 2
Response:	$\text{grade}^{3/2}$	status	retained
<i>N</i>	5797	5840	5863
Deviance	72878.5	4690.6	3906.5
Overall significance	$\chi^2_{16} = 3,976.6$ $P < 0.001$	$\chi^2_6 = 1,340.4$ $P < 0.001$	$\chi^2_{10} = 239.9$ $P < 0.001$
Fixed terms: <i>P</i> -values for model selection tests*			
P_CONDIT	< 0.001	0.362	< 0.001
P_ENGAGE	< 0.001	0.440	0.149
P_OPPORT	< 0.001	0.054	0.173
pol(SPWA, 2)	< 0.001	< 0.001	NE
SEM_NUM	0.009	0.198	NE
ATAR	< 0.001	< 0.001	0.023
GENDER	0.449	0.657	0.004
INDIGINEOUS	0.531	NE	NE
SES	0.163	NE	NE
REMOTE	0.603	NE	0.009
REGIONAL	0.218	NE	0.285
PARENT	0.801	NE	NE
AGE	0.918	0.556	0.001
ATTEND	< 0.001	< 0.001	0.036
ATTEND $\times$ P_ENGAGE	0.002	0.126	0.009
ATTEND $\times$ GENDER	0.003	NE	0.140
ATTEND $\times$ INDIGINEOUS	0.555	NE	NE
ATTEND $\times$ SES	0.929	NE	NE
ATTEND $\times$ REMOTE	0.203	NE	0.181
ATTEND $\times$ REGIONAL	0.587	NE	NE
ATTEND $\times$ PARENT	0.436	NE	NE
ATTEND $\times$ AGE	0.002	NE	NE
Random terms: estimated variance components			
δ ($\text{var}[e_i] = \sigma^2 \hat{y}_i ^{2\delta}$)	−0.31	—	—
(i) δ estimated			
COURSE	1762.2	—	—
UNIT	4476.7	—	—
Residuals (Dispersion)	717583	—	—
(ii) $\delta = 0$			
COURSE	1644.2	0.18	0.28
UNIT	4455.8	0.41	0.10
Residuals (Dispersion)	16746.2	—	—
COURSE ICC	0.072	0.046**	0.074**
UNIT ICC	0.195	0.105**	0.028**

Although diagnostic assessments demonstrated no evidence of any strong violations of model assumptions, the Q-Q plot of the estimates of realized values for UNIT (empirical Best Linear Unbiased Predictors: Henderson, 1950; Robinson, 1991) suggested that positive estimates were larger than expected under a normal distribution. Therefore, this analysis was repeated without UNIT, and similar results were found (see S2; Supplementary Material, for details). Although tests of UNIT and COURSE were statistically significant, estimates for ICCs showed that variation between levels of UNIT or COURSE were relatively low compared to the variation within those levels (see Table 2).

Tests for retained interaction terms were significant for ATTEND $\times$ AGE ($\chi^2 = 10.6$, $df = 1$, $P = 0.001$), ATTEND $\times$ GENDER ($\chi^2 = 9.6$, $df = 1$, $P = 0.002$), and ATTEND $\times$ P_ENGAGE ($\chi^2 = 9.7$, $df = 1$, $P = 0.002$). A positive coefficient for ATTEND $\times$ GENDER ($\hat{\beta} = 4.65 \pm 1.45$ SE) revealed that the slope of the linear relationship of transformed grade with ATTEND was higher for males than for females (reference level). Plots of back-transformed marginal means of grade, with 95% confidence intervals, demonstrate the nature and size of these effects for each gender, after controlling for the influence of other variables in the final model (Fig. 1). The model-predicted grade for males who had attended 11 sessions was 71.5% (95% CI: 68.2–74.7) as compared with 61.1% (58.8–63.4) for zero sessions attended, which was an average +10.4% gain from full SI attendance. Model predicted grades for females who had attended 11 sessions was 67.9% (64.9–70.8) compared with 61.7% (59.3–63.9) for zero sessions attended, which was an average +6.2% gain.

A negative coefficient for ATTEND $\times$ P_ENGAGE ($\hat{\beta} = -5.15 \pm 1.72$ SE) revealed that the slope of the linear relationship of transformed grade with ATTEND was lower for students who had prior experience with SI than for students who had not (reference level). Plots of back-transformed marginal means of grade, with 95% confidence intervals, demonstrate the nature and size of these effects for students with and without prior engagement (Fig. 2). The model-predicted grade for students with no prior engagement who had attended 11 sessions was 70.2% (95% CI: 67.0–73.4) as compared with 61.2% (58.9–63.5) for zero sessions attended, which was an average +9.0% gain from full SI attendance. Model predicted grades for students with some prior engagement was 66.4% (63.6–69.3) compared with 62.0% (59.7–64.4) for zero sessions, which was an average +4.4% gain.

A significant ATTEND $\times$ AGE interaction indicates that the slope of the linear relationship of transformed grade on ATTEND changes with the value of AGE. The negative coefficient for the ATTEND $\times$ AGE

interaction ($\hat{\beta} = -0.25 \pm 0.08$ SE) is reflected in the plotted model-predicted marginal means with ATTEND for each of four student ages: 18 years, 28 years, 38 years and 48 years (Fig. 3). This shows that the average grade increased with number of sessions attended and that the net increase (e.g., between zero and 11 sessions; Fig. 3) was higher for younger students than for older students. However, we refrain from quantifying these net effects further, as there were relatively few data points for some combinations of student age (especially for the older-aged students) and number of sessions attended.

2.2. Status (Pass/Fail): GLMM 1

Retained predictor variables in the final selected model, together with summary statistics and estimated variance components are shown in Table 2. Satisfactory convergence could not be achieved for the fit of the loaded mean model to these data because of overfitting (complete separation). Therefore, several candidate model terms (and predictor variables) were removed until loaded mean model convergence was achieved (Table 2). The subsequent top-down procedure selected the polynomial transform of previous semester weighted average, ATAR score, and the main effect of ATTEND as terms to be retained in the final model. Although tests of UNIT and COURSE were statistically significant, estimates for ICCs showed that variation between levels of UNIT or COURSE were relatively low compared to the variation within those levels (Table 2).

The χ^2 difference test for ATTEND was significant ($\chi^2 = 50.4$, $df = 1$, $P < 0.001$). This indicates that SI attendance had a significant effect on the proportion of students passing, after controlling for the other variables in the final model. A positive coefficient for ATTEND ($\hat{\beta} = 0.18 \pm 0.03$ SE) means that passing was more likely with an increasing number of SI sessions attended, at the scale of log odds (Fig. 4). The average probability of passing increases with number of sessions attended, but the increase from each additional session decreases as the number of attended sessions increases (Fig. 4). The model-predicted probability of passing for students who had attended 11 sessions was 0.98 (95% CI: 0.96–0.99) as compared with 0.88 (0.84–0.91) for zero sessions attended, which was an average +0.10 gain from complete SI attendance.

2.3. Retention: GLMM 2

Satisfactory convergence could not be achieved for the fit of the

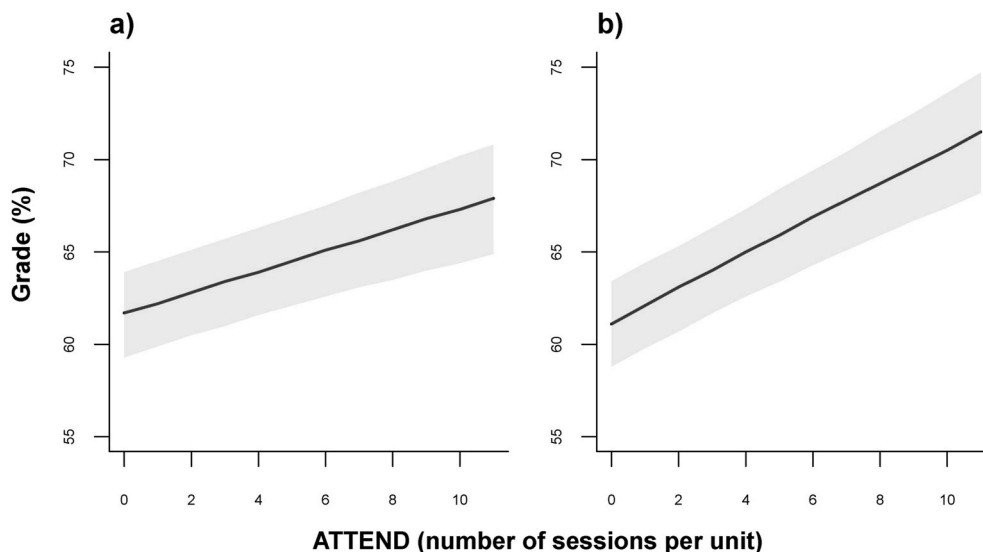


Fig. 1. Back-transformed marginal means of grade with ATTEND for a) GENDER = Females; b) GENDER = Males. 95% confidence interval indicated by shaded region.

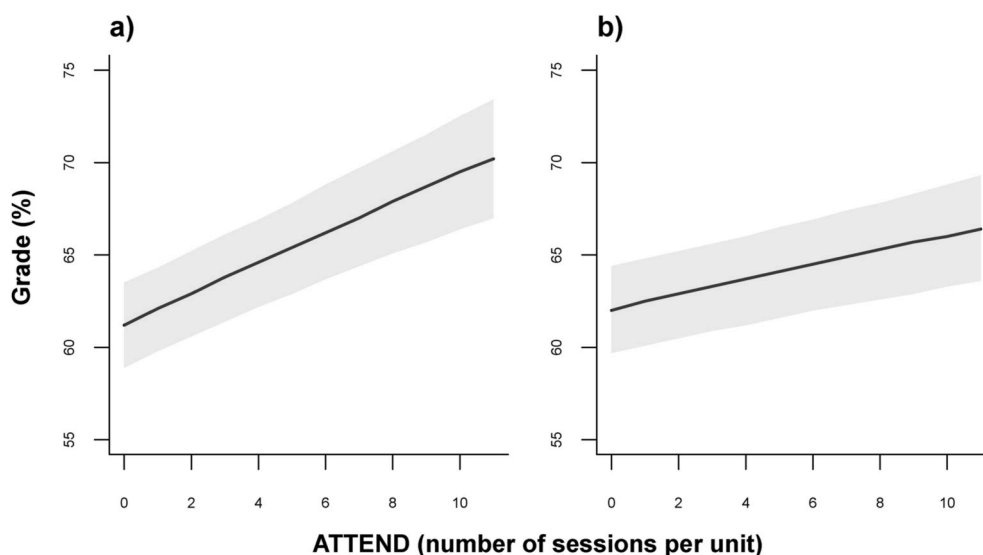


Fig. 2. Back-transformed marginal means of grade with ATTEND for a) P_ENGAGE = No; b) P_ENGAGE = Yes. 95% confidence interval indicated by shaded region.

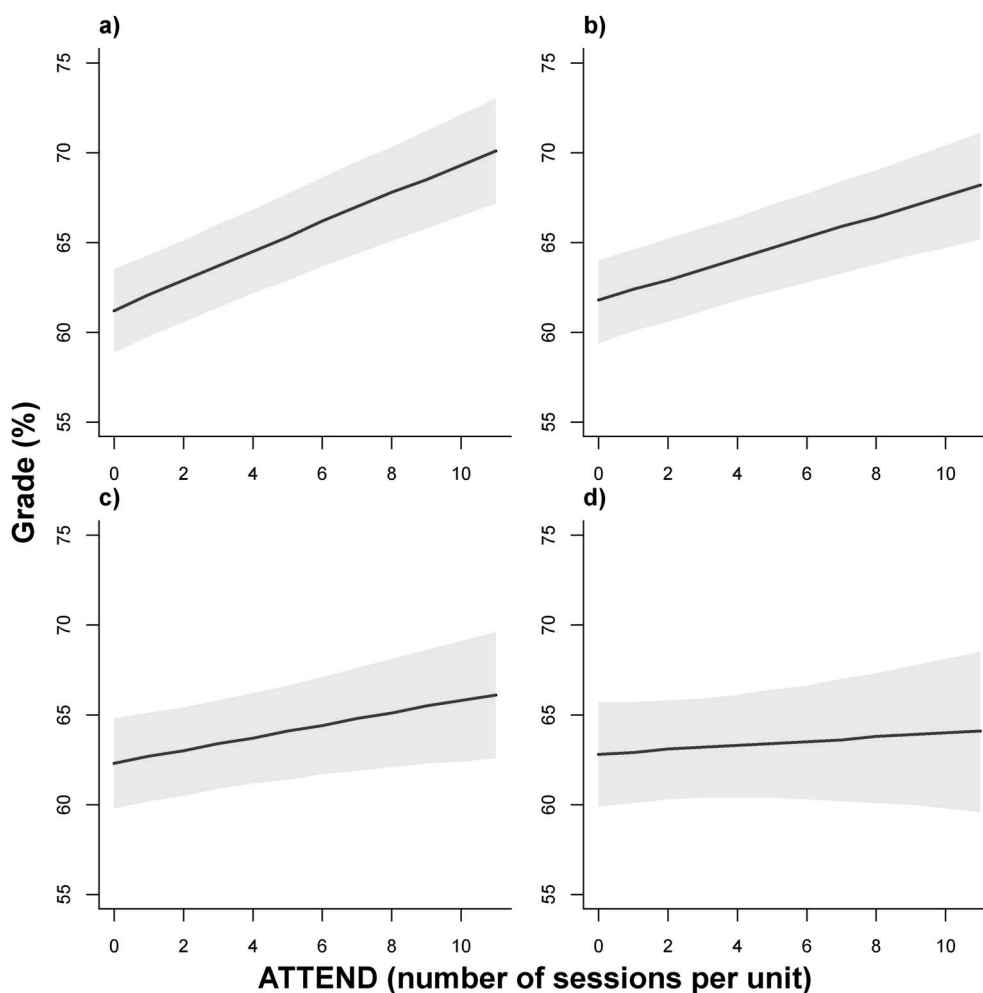


Fig. 3. Back-transformed marginal means of grade with ATTEND for a specific student age: a) 18 years; b) 28 years; c) 38 years; d) 48 years. 95% confidence interval indicated by shaded region.

loaded mean model due to overfitting, so top-down model selection commenced with a simplified model that excluded SPWA, SEM_NUM, INDIGENOUS, SES, PARENT, and several of the interaction terms (Table 2). The subsequent top-down procedure selected P_CONDIT,

P_ENGAGE, ATAR, GENDER, REMOTE, AGE, ATTEND, and ATTEND $\times$ P_ENGAGE as terms to be retained in the final model (GLMM 2; Table 2). Although tests of UNIT and COURSE were statistically significant, estimates for ICCs showed that variation between levels of UNIT or COURSE

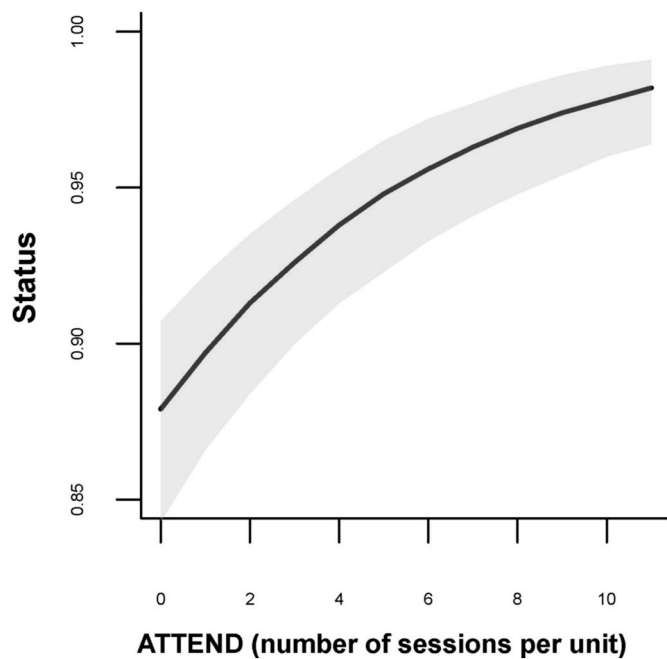


Fig. 4. Back-transformed marginal means of status with ATTEND. 95% confidence interval indicated by shaded region.

were relatively low compared to the variation within those levels.

The χ^2 difference test for the retained $\text{ATTEND} \times \text{P_ENGAGE}$ interaction was significant ($\chi^2 = 5.6$, $df = 1$, $P = 0.018$). A negative coefficient for this interaction term ($\hat{\beta} = -0.15 \pm 0.07 \text{ SE}$) indicates that the net change in proportion of students retained into the subsequent year, as a consequence of attending 11 sessions, was higher on average for students who had no prior experience with the program (reference level), as compared to those who had. Plots of the back-transformed model-predicted values by P_ENGAGE category demonstrate the nature of this interaction (Fig. 5). Attending a higher number of SI sessions increases the probability of retention into the subsequent year for both groups, in a non-linear fashion. The model-predicted probability of retention for students with no prior engagement who had attended 11 sessions was 0.99 (95% CI: 0.96–1.00) as compared with 0.90 (0.87–0.92) for zero sessions attended, which was an average

+0.09 gain from complete SI attendance. The model-predicted probability of retention for students with some prior engagement who had attended 11 sessions was 0.96 (0.92–0.98) as compared with 0.93 (0.90–0.95) for zero sessions attended, which was an average +0.03 gain.

3. Discussion

In this study, a multivariable analytic approach, with data collected from over 4000 students over a period of three years, was used to assess the effectiveness of an SI program at a large Australian university. We found that SI attendance was significantly and positively associated with students' grades, as well as their odds of passing and retention into the next academic year. Specifically, the subject grades of male students who attended a full semester of SI exceeded those of male students who did not attend by an estimated 10.4%. For female students, the difference was 6.2%. For students who had never before engaged with the program, a full semester of SI attendance was associated with a 9.0% grade increase. For those with some prior engagement, the increase was 4.4%. Those students who were youngest experienced largest grade increases. Furthermore, full attendees were 10% more likely to pass their subjects than non-attendees. Finally, for students who had never before engaged with the program, a full semester of SI attendance was associated with a 9.0% increase in the probability of reenrolling the following year. For students who had engaged previously, the increase in the probability of reenrolling was a more modest 3.0%. The use of linear (for grades) and generalized linear (for pass rates and retention) mixed models allowed us to account for hierarchical structure in our data (related observations for students enrolled in the same units and courses). This is something that is commonly encountered, but less commonly addressed in educational research (Murray & Hannan, 1990). These analyses also allowed us to control for the influence of prior ability, achievement and motivation, as well as several demographic variables. Including these terms in the models means that our adjusted estimates are less likely to be confounded by such sources of variation, which might otherwise have been accounted for more explicitly in a suitable experimental design.

When combined with the results from other large-scale studies (e.g., Dancer et al., 2007, 2015; Kochenour et al., 1997; Paloyo et al., 2016; van der Meer et al., 2017), the results of the current study demonstrate, quite unambiguously, that SI works. Despite this, relatively few students engage with typical SI programs, which are almost always voluntary. For

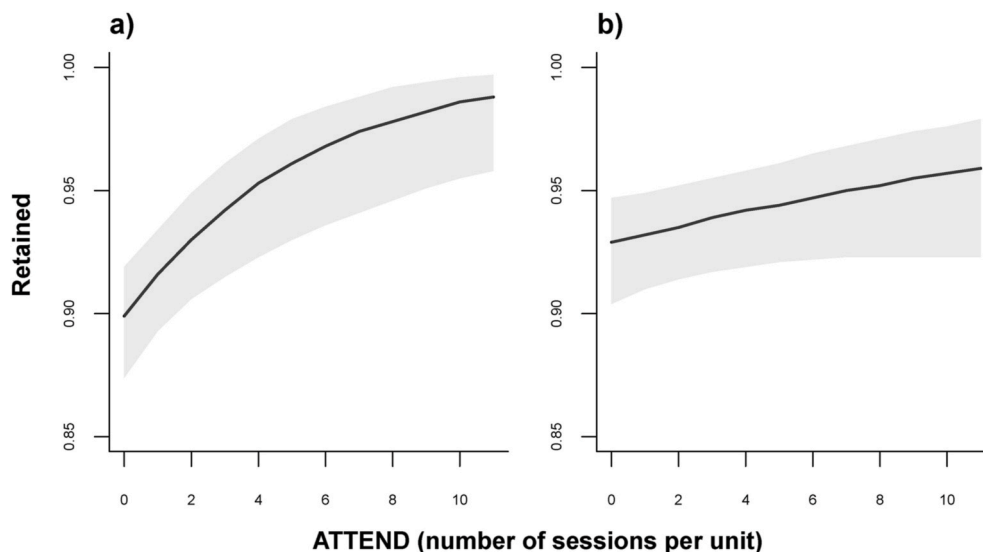


Fig. 5. Back-transformed marginal means of retained with ATTEND for a) P_ENGAGE = No; b) P_ENGAGE = Yes. 95% confidence interval indicated by shaded region.

example, in [Dancer et al. \(2007\)](#), less than a third of students attended one or more sessions, and only 7% attended 10 or more. In a follow-up study several years later, the number of students attending one or more sessions was even lower, at 20% ([Dancer et al., 2015](#)). In [van der Meer et al. \(2017\)](#) the SI program increased in popularity over time. However, by its fourth year of operation it was still only engaging 14.9% of eligible students. In [Kochenour et al. \(1997\)](#) and [Paloyo et al. \(2016\)](#), students attended an average of 1.26 of 10 sessions and 2.67 of 12 sessions respectively. In both studies, the majority of students did not attend a single SI session. Future researchers need to prioritize this issue, rather than the issue of effectiveness. In this regard, the limited available evidence suggests that psychological factors (e.g., [Allen, Tonta, Pereira, & Roberts, 2019](#)) may warrant greater consideration than socio-demographic factors or academic abilities (e.g., [van der Meer et al., 2017](#)).

In terms of students' grades, some interaction terms that were found to be significant should be discussed. Notably, students with no prior engagement in SI demonstrated greater average improvements in grades from participating, as compared to students who had some prior experience. One might be tempted to interpret this result to suggest that if universities target students without prior engagement to participate, this may lead to larger improvements in learning outcomes, on average, for the enrolled student population. However, such an inference would not be supported from the presented evidence, because we do not have measurements of the responsiveness to SI by students who did not attend. Furthermore, while this study also found greater improvements in average grades with increasing attendance estimated for male students as compared to those estimated for female students, these findings should be treated cautiously until replicated. Nonetheless, the emphasis of the program upon particular collaborative and communication skills indicates that improvement rates may be higher with particular individuals or groups. Further research ought to explore these nuances, and whether certain aspects of SI are better suited to different subgroups of students, notwithstanding the benefits for general skills development that all students would gain.

This study's main findings, that full attendance is associated with grade increases of up to 10.4%, and that full attendees are 10% more likely to pass their subjects and up to 9% more likely to continue their studies into the following year have significant implications for university management. For instance, given the financial implications associated with student performance and retention ([de Freitas et al., 2015](#)), there are significant benefits that can arise from the implementation of SI programs. Following from these findings, questions inevitably may be asked such as whether the SI approach be embedded into broader formal university study, and whether it would be as effective in this manner (i.e., beyond its use as a co-curricular form of study support). That is, it may be worthwhile to consider whether there are unique SI components that could be embedded within pedagogical design more broadly (or, even more generally, to explore why SI appears to be so effective). Indeed, in this light, possible linkages between cognitive load reduction/management may be worth considering (c.f., [Martin & Evans, 2018](#)). Follow-on questions may then also be asked, such as whether the program should be mandatory for all students, and in which manner this would be most effective.

Alongside these specific SI-related questions and implications, the present study also highlights the potential use and extension of a multivariable analytic approach for evaluation and benchmarking of programs. In this manner, such an approach provides a valuable mechanism for universities in their attempts to track the effectiveness of their own programs. We believe that these statistical possibilities ought to encourage educators/university managers and data analysts to collaborate in order to monitor and make iterative improvements to the way that their learning/pedagogical programs are run and amended over time. Such partnerships would likely help to overcome some of the

problems that frequently affect educational research (i.e., the lack of analyzing hierarchical data and sufficiently controlling for confounds at various levels), and thus lead to optimized outputs over time (both in terms of scholarly literature and practical learning programs/systems).

Importantly, there are some additional caveats to note when interpreting these findings. First, although this analysis investigated effects of proxies for prior motivation (P_ENGAGE, P_OPPORT), it did not deal with 'motivation' (self-reported or otherwise) at the time SI was undertaken. Second, the effect of SI was not assessed for students attending SI for the first time because data from the first semester of attendance in the program for each student were used to calculate prior engagement and motivation variables. Third, this study assessed the effectiveness of a voluntary program and therefore predictions concerning more general effects on student learning outcomes cannot be made. Fourth, although the ATAR score was shown to be important for explaining outcome measures of supplemental instruction, these data were available for domestic students who entered through the traditional school leaver route only. Thus, the models presented in this paper do not speak to the impact of the SI program for students who entered higher education via non-traditional (e.g., successful completion of an enabling program) pathways, or as international students. However, it should be noted that additional analyses including the latter cohort are reported in Supplementary Material S3. In these analyses there is no evidence to suggest that the effects of attendance differed between domestic and international students. Fifth, we were unable to incorporate any information about the personal characteristics or effectiveness of the UniPASS facilitators into our analyses, and thus the extent to which variability between facilitators impacted our results, or student attendance, is unknown. One final limitation is that the presented effects of SI attendance were estimated from cross-sectional data. An improved analysis would ideally focus on longitudinal data, whereby the time series of changes in student learning outcomes with subsequent sessions attended could be assessed for each student (and on aggregate, for a cohort of students, within each semester).

In conclusion, in this study we used a multivariable analytic approach to demonstrate how percentage grade increases of up to 10.4% were associated with a full semester of SI attendance. Furthermore, full attendees were 10% more likely than non-attendees to pass their subjects, and up to 9% more likely to continue their studies into the following year. These effects persisted after controlling for prior ability, achievement and motivation, as well as several demographic factors. The analytic approach used in this study can serve as an example for how existing datasets can be used to conduct a rigorous evaluation of the effectiveness of an educational intervention. A key benefit could be in the timely provision of results to inform subsequent planning decisions, for possible expansion or contraction of the program, or to target specific subjects or groups of students, such as first year enrolments. Given that the effectiveness of SI has been consistently demonstrated, future studies ought to consider other key questions. Most pressing amongst these is the question of why so few students engage with an academic support program that is so demonstrably effective.

CRediT authorship contribution statement

Peter J. Allen: Conceptualization, Methodology, Investigation, Writing – original draft, Writing – review & editing. **Sara de Freitas:** Conceptualization, Writing – original draft, Writing – review & editing. **Ross J. Marriott:** Conceptualization, Methodology, Formal analysis, Data curation, Visualization, Writing – original draft, Writing – review & editing. **Raphael M. Pereira:** Conceptualization, Investigation, Funding acquisition, Writing – original draft. **Cameron Williams:** Conceptualization, Writing – original draft. **Christopher J. Cunningham:** Conceptualization, Writing – original draft. **Douglas Fletcher:** Conceptualization.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.learninstruc.2021.101481>.

Appendix

Description of Variables Used in Analyses

Name	Description
grade	Final mark (0–100%) for a unit completed by a student within a semester.
status	Whether the student passed or failed the unit (binary).
retained	Whether the student's enrolment continued into the subsequent year or not (binary).
ATTEND	Number of SI sessions attended by the student for a specific unit in a semester.
SEM_NUM	Semesters enrolled since first studied a SI subject. Used as a time-specific variable for each student, controlling for years, and semesters within each year.
P.CONDIT	Prior conditional status: has the student been placed on conditional status (which is typically due to failing a unit) for any previous semester: Yes/No.
P.ENGAGE	Prior engagement: has the student previously participated in SI: Yes/No.
P.OPPORT	Prior opportunities taken: Average SI attendance (number of sessions) per unit per semester in previous semesters.
SPWA	Weighted average unit grade (0–100) for previous semester of university study.
ATAR	Standardized tertiary entrance score (0–100) for Traditional pathway students. A measure prior academic ability.
GENDER	Gender.
INDIGINEOUS	Indigenous. Was the student of Aboriginal or Torres Strait Islander origin? (Yes/No).
SES	Low socioeconomic status: Yes/No.
REMOTE	Remote. Was the student from a remote area: Yes/No.
REGIONAL	Regional. Was the student from a regional area: Yes/No.
PARENT	Number of parents with a tertiary qualification: {0, 1, 2}
AGE	Student age (years)
STUDENT_ID	Student ID: Nested within levels of UNIT and COURSE
UNIT	Unit ID: Partially crossed with COURSE. $N = 61$
COURSE	Course ID: Partially crossed with UNIT. $N = 35$

References

- Allen, P. J., Tonta, K. E., Pereira, R. M., & Roberts, L. D. (2019). Predicting peer-assisted study session attendance. *Active Learning in Higher Education*, 20(3), 249–262. <https://doi.org/10.1177/1469787417735613>
- Australasian Centre for PASS. (2010). Peer assisted study sessions (PASS) - guidelines for best practice. Retrieved from <http://www.uow.edu.au/content/groups/public/@web/@stsv/@pass/documents/doc/uow099559.pdf>.
- Australasian Centre for PASS. (2017). PASS trained institutions. Retrieved from <https://www.uow.edu.au/engage/pass-centre/training/institutions/>.
- Australian Government Department of Education and Training. (2017). Improving retention, completion and success in higher education: Higher Education Standards Panel. discussion paper, June 2017. Canberra, Australia: Author. Retrieved from https://docs.education.gov.au/system/files/doc/other/final_discussion_paper.pdf.
- Bates, D., & Martin, M. (2015). Fitting linear mixed-effects models using lme4. *Journal of Statistical Software*, 67(1), 1–48. <https://doi.org/10.18637/jss.v067.i01>
- Cheng, D., & Walters, M. (2009). Peer-assisted learning in mathematics: An observational study of student success. *Journal of Peer Learning*, 2(3), 23–39. <https://ro.uow.edu.au/ajpl/vol2/iss1/3/>.
- Curtin University. UniPASS study sessions. Retrieved from <https://students.curtin.edu.au/study-support/unipass/>, 2017.
- Dancer, D., Morrison, K., & Smith, M. (2007). Measuring the impact of a peer assisted learning program on students' academic performance in econometrics. In M. Davies, & R. Lever (Eds.), *The quantitative analysis of teaching and learning in higher education: Forum proceedings*. Melbourne, Australia: University of Melbourne. https://espace.library.uq.edu.au/data/UQ_2405565/flexiblelearning.pdf?Expires=1616778797&Key-Pair-Id=APKAJKNBJ4MJJC6NLQ&Signature=bf1n4qIGzhspovHVyQbheDJNI09Lcy4-h5mW5qWP-jvuW092t08z2CGCWEix31P0CPCmdtHwY6NkkuOZq2Tr3OZ5EKY4hJ5vAzYz6sXwn5t5Jnlzy5u9obiOf6uqCvWYC4Z7KgV3TdxK72-81lscUBbwMjPvV3YFYadCpCDfe2a9W0HJXUZWWlzpml2V5DVU4dZnflp2p9TEUGFDhMirG3HTK-Vw5P-EjYqOaQRrOV7hGUsr0EN-w55YjZ3mDhmY9sexGUwB0777JnACcKrHw8SQK-q5euZNVZLmnellxyF6mf3C3xgq1YtdRyhWtZKJdbC382BssZRkapj33sg.
- Dancer, D., Morrison, K., & Tarr, G. (2015). Measuring the effects of peer learning on students' academic achievement in first-year business statistics. *Studies in Higher Education*, 40(10), 1808–1828. <https://doi.org/10.1080/03075079.2014.916671>
- Dawson, P., van der Meer, J., Skaliky, J., & Cowley, K. (2014). On the effectiveness of supplemental instruction: A systematic review of supplemental instruction and peer-assisted study sessions literature between 2001 and 2010. *Review of Educational Research*, 84(4), 609–639. <https://doi.org/10.3102/0034654314540007>
- DeBerard, M. S., Spielmanns, G., & Julka, D. (2004). Predictors of academic achievement and retention among college freshmen: A longitudinal study. *College Student Journal*, 38(1), 66–80. <https://eric.ed.gov/?id=EJ701984>.
- Fox, J. (2003). Effect displays in R for generalised linear models. *Journal of Statistical Software*, 8(15), 1–27. <https://doi.org/10.18637/jss.v008.i15>
- Fox, J., & Weisberg, S. (2011). *An R companion to applied regression* (2nd ed.). Thousand Oaks CA: Sage.
- de Freitas, S., Gibson, D., Du Plessis, C., Halloran, P., Williams, E., Ambrose, M., ... Arnab, S. (2015). Foundations of dynamic learning analytics: Using university student data to increase retention. *British Journal of Educational Technology*, 46(6), 1175–1188. <https://doi.org/10.1111/bjet.12212>
- Henderson, C. R. (1950). Estimation of genetic parameters. *The Annals of Mathematical Statistics*, 21(2), 309–310. <https://www.jstor.org/stable/2236913>.
- International Center for Supplemental Instruction. (2021). Accreditation overview. Retrieved from <http://info.umkc.edu/si/certification/>.
- Kochenour, E. O., Jolley, D. S., Kaup, J. G., Patrick, D. L., Roach, K. D., & Wenzler, L. A. (1997). Supplemental instruction: An effective component of student affairs programming. *Journal of College Student Development*, 38, 577–586. <https://eric.ed.gov/?id=EJ557886>.
- MacCallum, R. C., Zhang, S., Preacher, K. J., & Rucker, D. D. (2002). On the practice of dichotomization of quantitative variables. *Psychological Methods*, 7(1), 19. <https://doi.org/10.1037/1082-989X.7.1.19>
- Martin, D. C. (2008). Foreword. *Journal of Peer Learning*, 1, 3–5. <https://ro.uow.edu.au/ajpl/vol1/iss1/2/>.
- Martin, A. J., & Evans, P. (2018). Load reduction instruction: Exploring a framework that assesses explicit instruction through to independent learning. *Teaching and Teacher Education*, 73, 203–214. <https://doi.org/10.1016/j.tate.2018.03.018>
- Martin, D. C., & Arendale, D. R. (1992). Supplemental instruction: Improving first year student success in high-risk courses: Monograph Series 7. Columbia, SC: National Resource Center for the Freshman Year Experience. <https://eric.ed.gov/?id=ED354839>.
- McCarthy, A., Smuts, B., & Cosser, M. (1997). Assessing the effectiveness of supplemental instruction: A critique and a case study. *Studies in Higher Education*, 22(2), 221–231. <https://doi.org/10.1080/03075079712331381054>
- van der Meer, J., Wass, R., Scott, S., & Kokaua, J. (2017). Entry characteristics and participation in a peer learning program as predictors of first-year students'

- achievement, retention, and degree completion. *AERA Open*, 3(3), 1–13. <https://doi.org/10.1177/2332858417731572>
- Murray, D. M., & Hannan, P. J. (1990). Planning for the appropriate analysis in school-based drug-use prevention studies. *Journal of Consulting and Clinical Psychology*, 58(4), 458–468. <https://doi.org/10.1037/0022-006X.58.4.458>
- Musah, R. A., & Ford, M. (2017). Peer-based supplemental instruction in STEM: Differences in effectiveness across transfer and nontransfer undergraduates. *Journal of Research on Educational Effectiveness*, 10(3), 596–618. <https://doi.org/10.1080/19345747.2016.1213341>
- Nieuwenhuis, R., te Grotenhuis, M., & Pelzer, B. (2012). influence.ME: Tools for detecting influential data in mixed effects models. *R Journal*, 4(2), 38–47. <https://doi.org/10.32614/RJ-2012-011>
- Paloyo, A. R. (2015). A note on evaluating supplemental instruction. *Journal of Peer Learning*, 8, 1–4. <https://ro.uow.edu.au/ajpl/vol8/iss1/2/>.
- Paloyo, A. R., Rogan, S., & Siminski, P. (2016). The effect of supplemental instruction on academic performance: An encouragement design experiment. *Economics of Education Review*, 55, 57–69. <https://doi.org/10.1016/j.econedurev.2016.08.005>
- Pinheiro, J., Bates, D., DebRoy, S., Sarkar, D., & R Core Team. (2015). nlme: Linear and nonlinear mixed effects models. *R package version 3.1-148*. <https://cran.r-project.org/web/packages/nlme/index.html>.
- R Core Team. (2015). *R: A language and environment for statistical computing*. Vienna, Austria: R Foundation for Statistical Computing. <https://www.r-project.org/>.
- Robinson, G. K. (1991). That BLUP is a good thing: The estimation of random effects. *Statistical Science*, 6(1), 15–32. <https://doi.org/10.1214/ss/1177011926>
- Snijders, T. A. B., & Bosker, R. J. (1999). *Multilevel analysis: An introduction to basic and advanced multilevel modeling*. Thousand Oaks, CA: Sage.
- Thammasiri, D., Delen, D., Meesad, P., & Kasap, N. (2014). A critical assessment of imbalanced class distribution problem: The case of predicting freshmen student attrition. *Expert Systems with Applications*, 41(2), 321–330. <https://doi.org/10.1016/j.eswa.2013.07.046>
- West, B. T., Welch, K. B., & Galecki, A. T. (2007). *Linear mixed models: A practical guide using statistical software*. Boca Raton, FL: Chapman & Hall/CRC.